solver.js — evaluateSafetyCheckpoints(result, config, TE_conv)
POST-CONVERGENCE SANITY CHECKS

warnings = result.warnings || []

TE_conv > result.T2?

Yes

No

warnings.push('Approach constraint flagged: TE > T2.')

(result.T2 - TE_conv) > 2?

Yes

warnings.push('Approach constraint flagged: T2 - TE > 2C.')

No

peakConfig = clone config, T0=43C
inverter? RPM=rpmMax, PR=1.0
: constant-speed? PR=0.95

peakResult = solveThermalSystem(peakConfig, TE_conv)
→ see 03-tc-secant-loop.mermaid

peakResult.converged?

No

Yes

warnings.push('Peak heat load flagged: system cannot balance at 43C.')

recompute UA at nominal conditions
LMTD_val = lmtd(mixed T1, result.T2, result.T2 - 2) *** uses T2-2 as a stand-in for TE, not the real solved TE ***
→ lmtd() in 10-evaporator-helpers.mermaid

UA * LMTD_val < 1.15 * peakResult.heatLoads.totalLoad?

Yes

warnings.push('Evaporator lacks 15% physical safety margin at 43C ambient.')

No

return result (with warnings)

(index.js: entry point)

never calls

validateHeatLoad.js — validateHeatLoad(geom, temps, calculatedLoads)
*** STANDALONE — imported in index.js but NEVER actually invoked inside runThermoAnalysis in the current source. DEAD CODE — safe to delete or wire in, but not both. ***

errors=[]; warnings=[]

T0/TF/TR all numbers?

No

Yes

errors.push('Missing or invalid temperature inputs.')

TF >= TR?

Yes

errors.push('Freezer temp warmer than or equal to Refrigerator temp.')

No

TR >= T0?

No

Yes

errors.push('Refrigerator temp warmer than or equal to Ambient temp.')

for each of 5 critical insulation points (F-top/left/right, R-top/left):

value undefined?

Yes

No

errors.push('Missing insulation thickness for X.')

value < 30mm?

Yes

warnings.push('X insulation extremely thin, expect massive heat leaks.')

No

calculatedLoads provided?

No

errors.push('No calculated loads provided for validation.')

Yes

QF < 0?

Yes

errors.push('Freezer heat load is negative.')

No

QR < 0?

Yes

errors.push('Refrigerator heat load is negative.')

No

totalLoad = QF+QR+(QEV||0)

totalLoad > 200?

Yes

warnings.push('Total heat load extremely high, most compressors will fail to satisfy demand.')

No

temps.TE defined and TE >= temps.TF?

Yes

errors.push('Evaporator temperature must be lower than Freezer temperature.')

No

return {isValid: errors.length===0, errors, warnings}

(index.js: successful result)

thermoUI.js calls EnergyConsumption(result) here, using the successful output

solver.js — EnergyConsumption(result)
*** Called by thermoUI.js (UI layer) AFTER runThermoAnalysis() returns — NOT invoked internally within the solver pipeline itself ***

result.converged === false?

Yes

return NaN

No

activeCyclePower = (compPower+fanPower+pwbOn)*PR
offCyclePower = pwbOff*(1-PR)
dailyBaseEnergy_kWh = (active+off)*24/1000

actualDefrostInterval = timerPeriod_h/PR
defrostEventsPerDay = 24/actualDefrostInterval
dailyDefrostEnergy_kWh = (defOnMin/60)*defHeater*events/1000

return {EnergyConsumption_kWhDay: base+defrost, EnergyConsumption_kWhMonth: daily*30}